

Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE- Academic Year 2026-27

Java Programming – Lab Batch 2025-29

Lab Assignment No:- 20	
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Batch	**A-2**
Class	**Second Year**
Academic Year & Semester	**26-27 & 3rd Sem**
Date of Submission	2/09/2026
Title of Assignment:	Develop a Java application to establish connectivity with a database and perform INSERT, UPDATE, and DELETE operations.

Practice questions
 	
Output Screenshots Solve questions on given link and attach screen shot of leetcode with your login and o/p
	1. Create a Java application to manage employee records using CRUD operations
	
	import java.sql.*;
	
	import java.util.Scanner;
	
	public class EmployeeCRUD {
	  public static void main(String[] args) {
	  String url = "jdbc:mysql://localhost:3306/companydb";
	  String username = "root";

```
        String password = "root";


        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url,
username, password);
            Scanner sc = new Scanner(System.in);


            while (true) {
                System.out.println("\n1. Insert");
                System.out.println("2. Display");
                System.out.println("3. Update");
                System.out.println("4. Delete");
                System.out.println("5. Exit");
                System.out.print("Enter choice: ");
                int choice = sc.nextInt();


                if (choice == 1) {
                    System.out.print("Enter Employee ID: ");
                    int id = sc.nextInt();
                    sc.nextLine();


                    System.out.print("Enter Name: ");
                    String name = sc.nextLine();


                    System.out.print("Enter Department: ");
```

```
                String department = sc.nextLine();

                System.out.print("Enter Salary: ");
                double salary = sc.nextDouble();

                PreparedStatement ps = con.prepareStatement(
                   "INSERT INTO employee VALUES (?, ?, ?, ?)"
                );

                ps.setInt(1, id);
                ps.setString(2, name);
                ps.setString(3, department);
                ps.setDouble(4, salary);
                ps.executeUpdate();

                System.out.println("Employee inserted
successfully.");
            }

            else if (choice == 2) {
                Statement stmt = con.createStatement();
                ResultSet rs = stmt.executeQuery("SELECT * FROM
employee");

System.out.println("\nID\tName\tDepartment\tSalary");
```

```
                while (rs.next()) {
                    System.out.println(
                        rs.getInt("employee_id") + "\t" +
                        rs.getString("name") + "\t" +
                        rs.getString("department") + "\t\t" +
                        rs.getDouble("salary")
                    );
                }
            }

            else if (choice == 3) {
                System.out.print("Enter Employee ID: ");
                int id = sc.nextInt();

                System.out.print("Enter New Salary: ");
                double salary = sc.nextDouble();

                PreparedStatement ps = con.prepareStatement(
                    "UPDATE employee SET salary = ? WHERE
employee_id = ?"
                );

                ps.setDouble(1, salary);
                ps.setInt(2, id);
                ps.executeUpdate();
```

```
                System.out.println("Employee updated
successfully.");
            }

            else if (choice == 4) {
                System.out.print("Enter Employee ID: ");
                int id = sc.nextInt();

                PreparedStatement ps = con.prepareStatement(
                    "DELETE FROM employee WHERE employee_id = ?"
                );

                ps.setInt(1, id);
                ps.executeUpdate();

                System.out.println("Employee deleted successfully.");
            }

            else if (choice == 5) {
                System.out.println("Exiting...");
                break;
            }

            else {
                System.out.println("Invalid choice.");
            }
```

```
            }


            con.close();
            sc.close();


        } catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

2.Create a Java application to manage student records using CRUD operations. Perform Create,Read, Update, and Delete operations on student details such as roll number, name, course, and marks using a database

```
import java.sql.*;
import java.util.Scanner;


public class StudentCRUD {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String username = "root";
        String password = "root";


        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url,
```

```
username, password);
        Scanner sc = new Scanner(System.in);


        while (true) {
            System.out.println("\n===== Student Record
Management =====");
            System.out.println("1. Create Student");
            System.out.println("2. Read Students");
            System.out.println("3. Update Student");
            System.out.println("4. Delete Student");
            System.out.println("5. Exit");
            System.out.print("Enter choice: ");
            int choice = sc.nextInt();

            if (choice == 1) {
                System.out.print("Enter Roll No: ");
                int rollNo = sc.nextInt();
                sc.nextLine();

                System.out.print("Enter Name: ");
                String name = sc.nextLine();

                System.out.print("Enter Course: ");
                String course = sc.nextLine();

                System.out.print("Enter Marks: ");
```

```
                double marks = sc.nextDouble();

                PreparedStatement ps = con.prepareStatement(
                   "INSERT INTO student VALUES (?, ?, ?, ?)"
                );

                ps.setInt(1, rollNo);
                ps.setString(2, name);
                ps.setString(3, course);
                ps.setDouble(4, marks);
                ps.executeUpdate();

                System.out.println("Student record created
successfully.");
            }

            else if (choice == 2) {
                Statement stmt = con.createStatement();
                ResultSet rs = stmt.executeQuery("SELECT * FROM
student");

                System.out.println("\nRoll
No\tName\tCourse\tMarks");
                System.out.println("--------------------------------");

                while (rs.next()) {
                   System.out.println(
```

```
                    rs.getInt("roll_no") + "\t" +
                    rs.getString("name") + "\t" +
                    rs.getString("course") + "\t" +
                    rs.getDouble("marks")
                );
            }
        }

        else if (choice == 3) {
            System.out.print("Enter Roll No: ");
            int rollNo = sc.nextInt();

            System.out.print("Enter New Marks: ");
            double marks = sc.nextDouble();

            PreparedStatement ps = con.prepareStatement(
                "UPDATE student SET marks = ? WHERE roll_no = ?"
            );

            ps.setDouble(1, marks);
            ps.setInt(2, rollNo);
            ps.executeUpdate();

            System.out.println("Student record updated successfully.");
```

```
            }

            else if (choice == 4) {
                System.out.print("Enter Roll No: ");
                int rollNo = sc.nextInt();

                PreparedStatement ps = con.prepareStatement(
                    "DELETE FROM student WHERE roll_no = ?"
                );

                ps.setInt(1, rollNo);
                ps.executeUpdate();

                System.out.println("Student record deleted
successfully.");
            }

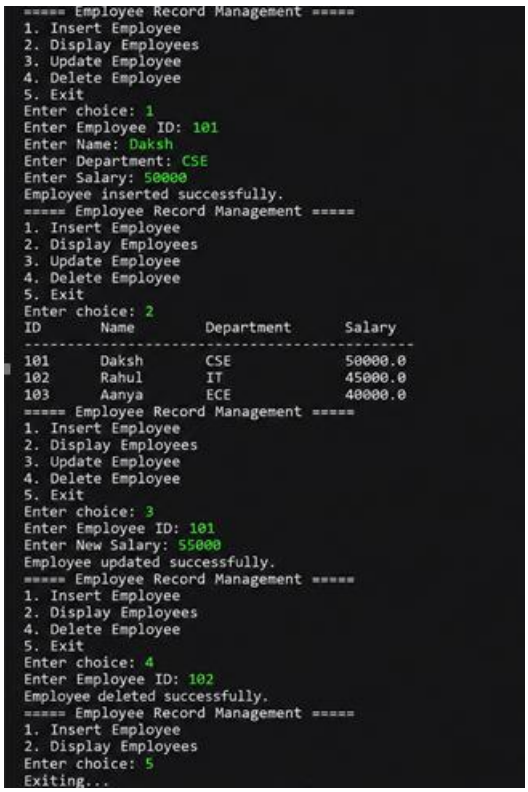
            else if (choice == 5) {
                System.out.println("Exiting...");
                break;
            }

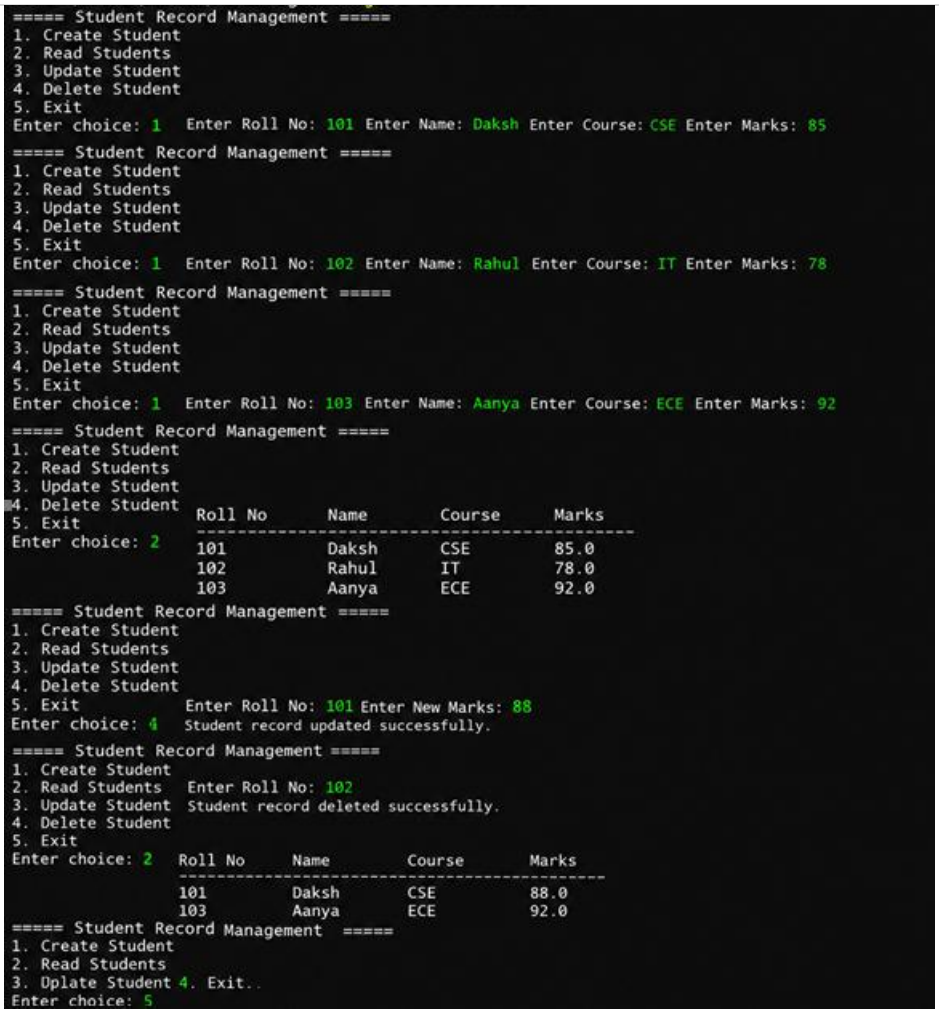
            else {
                System.out.println("Invalid choice.");
            }
        }
```

```
            con.close();
            sc.close();

        } catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

Output Screenshots (if applicable)





Conclusion	The program successfully manages student records using CRUD operations—Create, Read, Update, and Delete. JDBC is used to connect the Java application with the MySQL database and perform operations on student details such as roll number, name, course, and marks.